

Haiyan Zhao

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I am a third-year Computer Science PhD student at the New Jersey Institute of Technology, where I work with Prof. Mengnan Du on the **mechanistic interpretability of large language models**: how information is encoded in and decoded from their internal representations. This work has led to 8 first- and co-first-author papers at venues such as ICLR, ACL, EMNLP, and ACM TIST, spanning concept-subspace modeling, representation explanations, and sparse-autoencoder steering. I am now seeking a **research scientist internship** to extend these directions toward trustworthy AI and post-training.

Research Interests: Model Interpretability · Trustworthy AI · Large Language Models · Post-training

EDUCATION

New Jersey Institute of Technology <i>Ph.D. in Computer Science · Advisor: Dr. Mengnan Du</i>	2023 – Present Newark, NJ
Huazhong University of Science and Technology <i>M.E. in Control Engineering · Advisor: Dr. Xiaolong Shi</i>	2017 – 2019 Wuhan, China
Wuhan University of Technology <i>B.E. in Automation</i>	2013 – 2017 Wuhan, China

EXPERIENCE

Cloudflare <i>Research Scientist Intern</i>	2026.05 – 2026.08
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- Building tool-calling algorithms for LLM agents with dynamic activation-steering, extending interpretability methods to agent behavior control.

SELECTED PUBLICATIONS

Selected publications (* = equal contribution). Full list on [Google Scholar](#).

- [1] **H. Zhao**, Z. He, G. Wang, A. Payani, Y. Li, and M. Du, "Universal activation verbalizer: A unified framework for cross-model activation explanation," *arXiv preprint arXiv:2605.25903*, 2026. *Explain activations in natural language*.
- [2] **H. Zhao**, Z. He, F. Yang, A. Payani, and M. Du, "Rep2text: Decoding full text from a single llm token representation," *arXiv preprint arXiv:2511.06571*, 2025. *Reinverse activation into input text using adapters*.
- [3] J. He, **H. Zhao**, R. Shi, Y. Liu, X. Wang, F. Sun, and M. Du, "SAEExplainer: Interpreting sae features with activation-guided preference optimization," *arXiv preprint arXiv:2606.08496*, 2026. *Refines SAE-feature explanations via iterative DPO*.
- [4] Z. He, **H. Zhao**, A. Payani, and M. Du, "Memlens: Uncovering memorization in llms with activation trajectories," *arXiv preprint arXiv:2509.20909*, 2025. *Detects memorization from layer-wise activation trajectories*.
- [5] **H. Zhao**, H. Zhao, B. Shen, A. Payani, F. Yang, and M. Du, "Beyond single concept vector: Modeling concept subspace in llms with gaussian distribution," *ICLR*, 2025. *Models a concept as a Gaussian subspace to describe variance*.
- [6] **H. Zhao**, H. Chen, F. Yang, N. Liu, H. Deng, H. Cai, S. Wang, D. Yin, and M. Du, "Explainability for large language models: A survey," *ACM TIST*, vol. 15, no. 2, pp. 1–38, 2024. *A comprehensive, widely-cited survey of explainability for LLMs*.
- [7] **H. Zhao**, X. Wu, F. Yang, B. Shen, N. Liu, and M. Du, "Denoising concept vectors with sparse autoencoders for improved language model steering," *EACL Findings*, 2026. *Uses SAEs to denoise concept vectors*.
- [8] Z. He*, **H. Zhao***, Y. Qiao, F. Yang, A. Payani, J. Ma, and M. Du, "Saif: A sparse autoencoder framework for interpreting and steering instruction following of language models," *ECML PKDD*, 2026.
- [9] X. Wu*, **H. Zhao***, Y. Zhu*, Y. Shi*, F. Yang, T. Liu, X. Zhai, W. Yao, J. Li, M. Du, *et al.*, "Usable xai: 10 strategies towards exploiting explainability in the llm era," *TKDD*, 2026.
- [10] **H. Zhao**, F. Yang, S. Bo, H. Lakkaraju, and M. Du, "Towards uncovering how large language model works: An explainability perspective," *SIGKDD Explorations*, 2025.

TECHNICAL SKILLS

ML / LLM	PyTorch, HuggingFace Transformers & PEFT, vLLM, DeepSpeed / FSDP, Weights & Biases
Interpretability	Linear probing, Logit / Tuned Lens, Patchscopes, sparse autoencoders (SAEs), activation patching, representation steering, activation inversion
Post-training	Supervised fine-tuning, knowledge distillation, DPO